

Advanced Data Science Lab [MCA33PE21]

FA2 Case Study: Data Analysis Using Machine Learning & Streamlit App Deployment

Student Name	Shantanu Suryawanshi
PRN	125M1H026
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1. Dataset Selection

The selected dataset is the Student Performance dataset from the UCI Machine Learning Repository. The repository identifies it as a multivariate dataset suitable for both classification and regression. It contains demographic, social, school-related, study-related, and academic information collected from two Portuguese schools.

For this case study, the Mathematics course file, student-mat.csv, is used. The final grade G3 is treated as a continuous regression target. A binary Pass/Fail target is also derived from G3, with $G3 \geq 10$ treated as Pass and $G3 < 10$ treated as Fail.

Primary source: <https://archive.ics.uci.edu/dataset/320/student+performance>

2. Why This Dataset Fits the Case Study

Requirement	Fit
Regression	G3 is a continuous final grade, so Linear Regression is applicable.
Classification	G3 can be converted into a practical Pass/Fail target.
Preprocessing	The data contains numerical and categorical variables.
Feature Scaling	Scaling is useful for Logistic Regression, SVM and KNN.
EDA	Grades, absences, study time and educational variables provide meaningful relationships to analyze.
Streamlit	Ten selected features provide a manageable and understandable user interface.

3. Data Acquisition & Preprocessing

- The notebook downloads the official UCI student.zip archive and selects student-mat.csv.
- A public GitHub mirror is used only as a fallback if the UCI archive is temporarily unavailable.
- The first rows, shape, data types and missing-value counts are inspected.
- Ten relevant features are selected to keep the models interpretable and the Streamlit form practical.
- Numerical variables are median-imputed and standardized.
- Categorical variables are mode-imputed and one-hot encoded.
- An 80/20 train/test split is used. Classification uses stratification.
- Preprocessing is kept inside scikit-learn pipelines to prevent data leakage.

4. Exploratory Data Analysis

- Final-grade distribution is visualized to understand the target shape.
- G1 and G2 are compared with G3 because they represent earlier academic periods.
- Study time is compared with final grade using a box plot.
- A numerical correlation heatmap is used to identify relationships among quantitative variables.

5. Machine Learning Models

Model	Task	Evaluation
Linear Regression	Regression	MAE, MSE, R ²
Logistic Regression	Classification	Accuracy, Precision, Recall, F1
Decision Tree	Classification	Accuracy, Precision, Recall, F1
Random Forest	Classification	Accuracy, Precision, Recall, F1
Support Vector Machine	Classification	Accuracy, Precision, Recall, F1
K-Nearest Neighbors	Classification	Accuracy, Precision, Recall, F1
Naive Bayes	Classification	Accuracy, Precision, Recall, F1

6. Model Evaluation

The classification models are evaluated using accuracy, precision, recall and F1-score. Confusion matrices and classification reports are generated for each classifier. The regression model is evaluated using MAE, MSE and R^2 .

The notebook intentionally calculates the numerical scores from the downloaded dataset rather than hard-coding values into the report. This keeps the reported results traceable to the actual dataset used during execution.

7. Hyperparameter Tuning

GridSearchCV is applied to Random Forest using 5-fold cross-validation. The search covers the number of trees, maximum tree depth, and minimum samples required for a split. The best parameter combination and the best cross-validation accuracy are printed by the notebook. The tuned model is then evaluated on the untouched test set.

Parameter	Values searched
n_estimators	100, 150
max_depth	None, 5, 10
min_samples_split	2, 5
Cross-validation	5-fold
Scoring	Accuracy

8. Streamlit Deployment

- The Streamlit application is provided separately as app.py.
- The application uses the same dataset, selected features, preprocessing and model definitions as the notebook.
- If saved model files are not present, app.py trains the models and saves them with joblib.
- Subsequent runs load the saved pipelines instead of retraining unnecessarily.
- The interface supports final-grade regression and Pass/Fail classification.
- For classification, the user can select among Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Naive Bayes and tuned Random Forest.
- The application displays prediction results and relevant test-set model metrics.

9. Findings and Insights

- The dataset is well aligned with the case-study requirement because it supports both regression and classification.
- Earlier academic grades are highly informative for predicting the final grade; this relationship is also documented by the UCI dataset description.
- A single metric is not sufficient for classification, so precision, recall and F1-score are reported alongside accuracy.
- Pipeline-based preprocessing provides a consistent transformation for every model and avoids train/test leakage.
- GridSearchCV provides a systematic way to select Random Forest hyperparameters instead of relying only on default settings.
- The final Streamlit application converts the notebook workflow into an interactive prediction interface.

10. Submission Files

File	Purpose
125M1H026_ADSL_FA2_Case_Study.ipynb	Part 1 and Part 2 implementation, EDA, preprocessing, model evaluation and GridSearchCV
app.py	Streamlit web application for interactive prediction and model comparison.
125M1H026_ADSL_FA2_Case_Study_Report.pdf	Brief report covering dataset, methodology, models, tuning, findings and deployment.

Dataset citation: Cortez, P. (2008). Student Performance. UCI Machine Learning Repository. DOI: 10.24432/C5TG7T.

Note: Numerical model scores are generated directly by the notebook from the downloaded dataset; no performance values have been fabricated in this report.